Linear Regression Analysis

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1 Introduction

We will perform a linear regression for generated empirical data. The dependent variable Y is predicted based on two explanatory variables: X_1 and X_2. The linear relationship being investigated is expressed by the formula:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \varepsilon \tag{1}$$

where ε represents the random error term, and β_1 and β_2 are the parameters.

We generate a random sample $N = 100$ times using Wolfram Mathematica. The variables X_1 and X_2 are drawn from a uniform distribution over the interval $[0, 10]$. The explanatory variable vector X_1 and variable vector X_2 can be written as column vectors:

$$X_1 = \begin{bmatrix} x_{1,1} \\ x_{2,1} \\ \vdots \\ x_{100,1} \end{bmatrix}, \quad X_2 = \begin{bmatrix} x_{1,2} \\ x_{2,2} \\ \vdots \\ x_{100,2} \end{bmatrix} \tag{2}$$

These variables are combined into a single explanatory data matrix X with dimensions 100×2:

$$X = \begin{bmatrix} X_1 & X_2 \end{bmatrix} = \begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \\ \vdots & \vdots \\ x_{100,1} & x_{100,2} \end{bmatrix} \tag{3}$$

The random error term ε is generated from a normal distribution with an expected value of 0 and a standard deviation of 2.5, meaning $\varepsilon \sim \mathcal{N}(0, 2.5)$. The vector of Y values is calculated assuming the parameter values $\beta_1 = 3.5$ and $\beta_2 = -1.7$:

$$Y = 3.5 \cdot X_1 - 1.7 \cdot X_2 + \varepsilon = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{100} \end{bmatrix} \tag{4}$$

2 Model Parameter Estimation

To determine the estimators of the β parameters, we use the Ordinary Least Squares (OLS) method. In matrix notation, this estimator is determined using the pseudoinverse matrix, minimizing the sum of squared residuals:

$$\hat{\beta} = (X^T X)^{-1} X^T Y \tag{5}$$

As a result of the calculations in Mathematica, we obtained the following parameter estimators:

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 3.6254 \\ -1.8154 \end{bmatrix} \tag{6}$$

We can observe that the calculated values are close to the assumed values (3.5 and -1.7).

2.1 Model Fit in Three-Dimensional Space

The chart below presents the scatter plot of the 100 generated points in the three-dimensional space of variables (X_1, X_2, Y) along with the fitted linear regression plane defined by the calculated estimators $\hat{\beta}_1$ and $\hat{\beta}_2$.

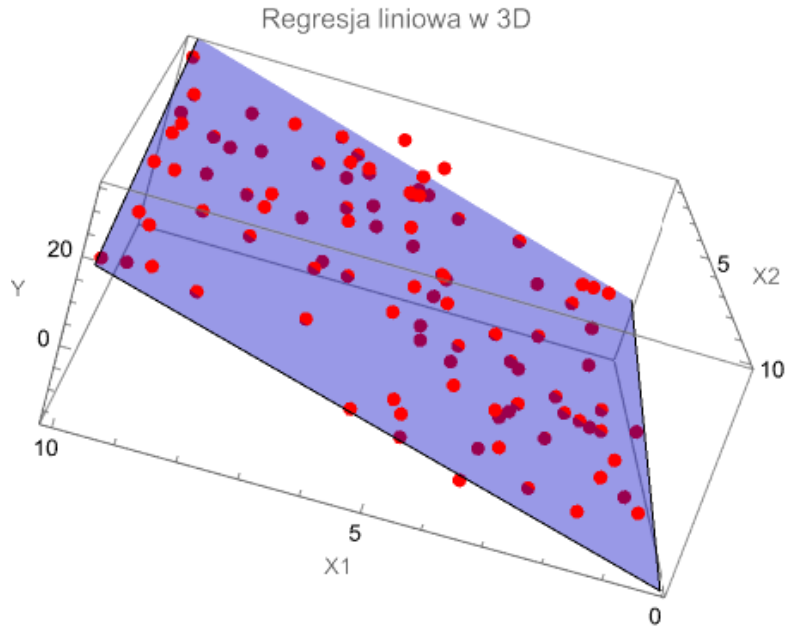


Figure 1: Three-dimensional scatter plot with the fitted regression plane.

3 Analysis of Explanatory Variables

An essential assumption in a regression model is the lack of correlation between explanatory variables. Using the software, we calculated the linear correlation coefficient for vectors X_1 and X_2, which was $r \approx -0.0338$. The value close to zero and the correlation chart below confirm that these variables are independent of each other.

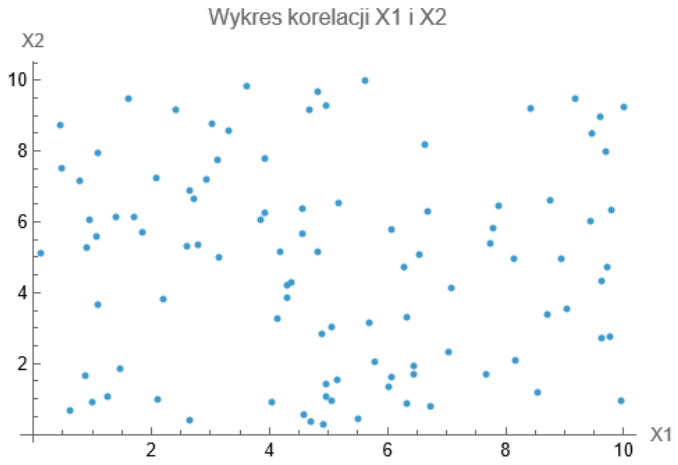


Figure 2: Scatter plot examining the correlation between variables X_1 and X_2.

4 Residual Analysis

Having calculated the parameter vector $\hat{\beta}$, we determined the prediction vector $\hat{Y}$ and the residual vector e:

$$\hat{Y} = X\hat{\beta} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{100} \end{bmatrix}, \quad e = Y - \hat{Y} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_{100} \end{bmatrix} \tag{7}$$

4.1 Distribution of Residuals

First, we examined the distribution of the calculated residual vector e. The Ordinary Least Squares method assumes that the residuals should have a distribution close to normal, as seen in the chart below.

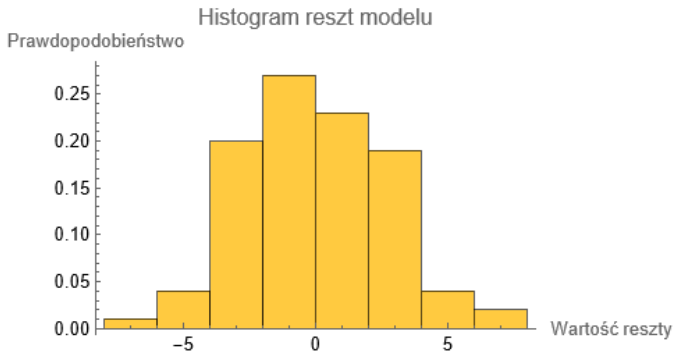


Figure 3: Histogram of the residual vector e.

4.2 Randomness of Residuals

Next, we analyzed the behavior of the residuals relative to the observation number i. The points in the chart below should be distributed randomly without forming any visible trends.

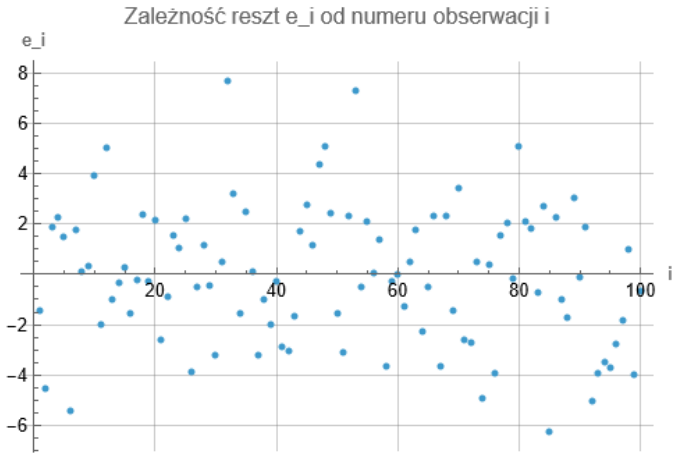


Figure 4: Dependence of residuals e_i on the observation number i.

Subsequently, we examined the dependence of the residuals e_i on the calculated fitted values $\hat{y}_i$ and presented it in the chart.

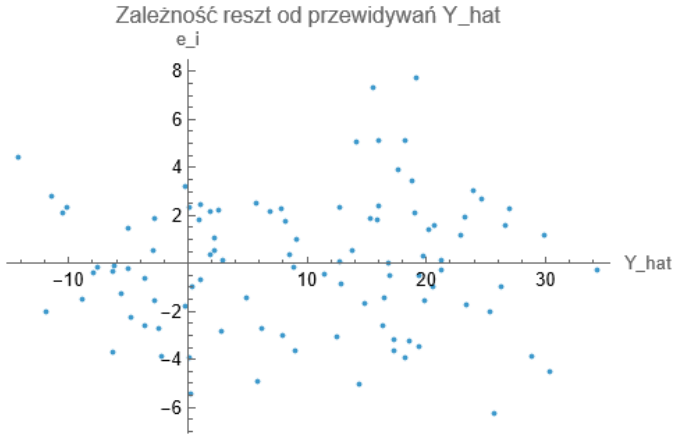


Figure 5: Dependence of residuals e_i on the fitted values $\hat{y}_i$.

4.3 Autocorrelation of Residuals

Finally, we conducted an autocorrelation analysis to check whether the error in a given step affects the error in the subsequent step. The dependence of the residual e_i on (e_{i+1}) and (e_{i+2}) is plotted on a single chart. The lack of visible correlation in the chart indicates that the model assumptions have been met.

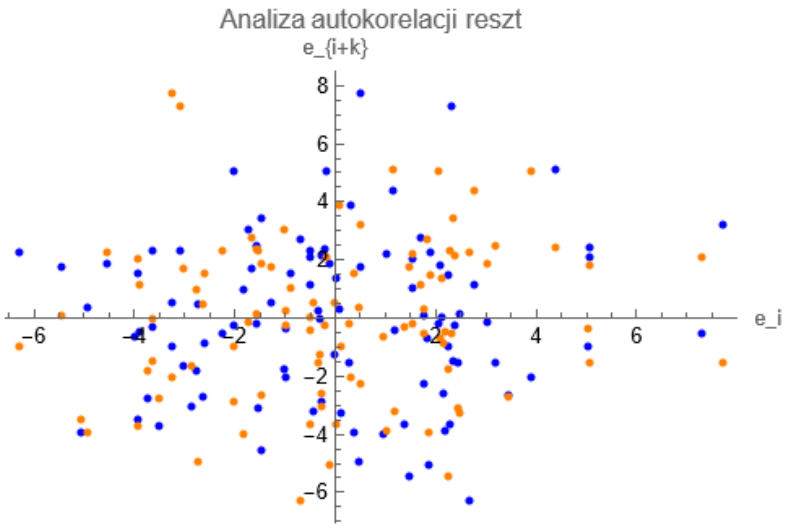


Figure 6: Autocorrelation check: dependence of e_i on e_{i+1} and e_{i+2}.

5 Conclusion

The parameter estimation performed in Wolfram Mathematica using the pseudoinverse matrix allowed for the determination of the regression plane. The analysis of the residual vector showed that the model is well-fitted, and the assumptions of the Ordinary Least Squares method were met (no autocorrelation, constant error variance, and a residual distribution close to normal).